

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/362746333>

Antimicrobial Resistance Genes, Virulence Genes, and Associated Mobile Genetic Elements of Eight Multidrug-Resistant Enterobacterales Isolated from Hospital-Acquired Urinary Tract...

Article in *Microbial drug resistance* (Larchmont, N.Y.) · August 2022

DOI: 10.1089/mdr.2022.0003

CITATIONS

4

7 authors, including:



Vindya Perera

Sabaragamuwa University of Sri Lanka

18 PUBLICATIONS 76 CITATIONS

SEE PROFILE



Nelun de Silva

Sabaragamuwa University of Sri Lanka

14 PUBLICATIONS 85 CITATIONS

SEE PROFILE

READS

116



Kushlani Jayatileke

Sri Jayewardenapura General Hospital, Sri Lanka

54 PUBLICATIONS 2,300 CITATIONS

SEE PROFILE



Virve I Enne

University College London

64 PUBLICATIONS 2,623 CITATIONS

SEE PROFILE



Antimicrobial Resistance Genes, Virulence Genes, and Associated Mobile Genetic Elements of Eight Multidrug-Resistant *Enterobacteriales* Isolated from Hospital-Acquired Urinary Tract Infections in Sri Lanka

Vindya Perera,^{1,2} Sara de Silva,³ Kushlani Jayatilleke,³ Nelun de Silva,^{2,4}
Alp Aydin,⁵ Virve Enne,⁵ and Enoka Corea¹

The study describes the first isolation of multidrug-resistant (MDR) *Klebsiella pneumoniae* ST16, *Escherichia coli* ST131 (Esc), and *Enterobacter hormaechei* subsp. *steigerwaltii* ST93 (*Enterobacter cloacae* complex [ECC]) in Sri Lanka. Eight MDR strains of uropathogenic *Enterobacteriales* isolated from hospital acquired urinary tract infections (UTIs) were analyzed using genomic sequencing and comparative genomics. Isolates carried multiple carbapenemase, AmpC, and ESBL (extended-spectrum β -lactamase) genes. ECC manifested both *bla*_{NDM-4} and *bla*_{OXA-181}. The *K. pneumoniae* strains harbored fimbrial genes that facilitate pathogenesis of UTI. Several extraintestinal pathogenic *E. coli* associated virulence genes were identified in Esc. The efflux pump gene, *acrA*, and the T6SS gene cluster were detected in ECC. Many antimicrobial resistance (AMR) and virulence genes were identified associated with mobile genetic elements. ISEcp1 flanked upstream of *bla*_{CTX-M-15}. The carbapenemase genes were carried on ColKP3 plasmids and were associated with ISEcp1. In Esc, the AMR gene *bla*_{TEM-1B} and virulence gene *traT* were found on an IncF plasmid replicon. In *K. pneumoniae* the AMR genes *sul1* and *tetB* present on IncR plasmid replicons and were associated with the insertion sequence IS6100. In Kp5, *bla*_{LAP-2} and *qnrS1* coexisted and were flanked by ISEcl. AMR gene clusters, conferring resistance to multiple antimicrobial classes, flanked by mobile elements were identified in seven isolates.

Keywords: antimicrobial resistance genes, virulence genes, mobile genetic elements, *Enterobacteriales*, hospital-acquired urinary tract infections, Sri Lanka

Background

THE SPREAD OF MULTIDRUG-RESISTANT (MDR) *Enterobacteriales* such as *Klebsiella pneumoniae*, *Escherichia coli*, and *Enterobacter cloacae* complex (ECC) strains resulting in treatment failure and increased mortality has become a global health problem.^{1,2} In particular, extended spectrum β -lactamase (ESBL) and carbapenemase-

producing strains of *K. pneumoniae*, *E. coli*, and ECC are of concern due to their association with transferable MDR and the establishment of successful clones.²⁻⁴ Such strains pose a potential threat to hospitals as they are known to cause hospital outbreaks.⁴

Several sequence types (STs) of MDR *K. pneumoniae*, such as ST14, 15, 16, 17, 23, 29, 101, and 147, are widely distributed across the globe.^{5,6} *E. coli* ST131 is notorious as

¹Department of Microbiology, Faculty of Medicine, University of Colombo, Colombo, Sri Lanka.

²Department of Microbiology, Faculty of Medicine, Sabaragamuwa University of Sri Lanka, Ratnapura, Sri Lanka.

³Sri Jayewardenepura General Hospital, Nugegoda, Sri Lanka.

⁴Neville Fernando Teaching Hospital, Malabe, Sri Lanka.

⁵Centre for Clinical Microbiology, Division of Infection and Immunity, University College London, Royal Free Campus, Rowland Hill Street, London, United Kingdom.

TABLE 1. CLINICAL DATA AND CHARACTERISTICS OF STRAINS UNDER STUDY

Molecular typing					Resistance profile CLSI disk diffusion												
Strain ID	Sample	Date	ST	Serotypes	AMR genes	Virulence genes	CPD	CAZ	CTX	CRO	AZM	FEP	FOX	CTT	IPM	MEM	ETP
Kp1	Urine	Oct-2015	ST16	K15, K17, K51, K52O3b	blaCTX-M-15, blaSHV-1, blaOXA-1, blaOXA-232, aac(6')-Ib-cr, erm(B), catB3, aac(6')-Ib-cr, oqxA, ARR-3, sul, dfrA1	fyuA, irp1, irp2, ybtA, ybtE, ybtP, ybtS, ybtT, ybtU, ybtX, traT, fimA, mrkA, mrkB, mrkC, mrkD, mrkF, mrkH, mrkI, mrkJ	R	R	R	R	R	R	R	R	R	R	R
Kp2	Urine	Oct-2015	ST147	K64 O2v1	blaCTX-M-15, blaSHV-11, blaTEM-1B, aph (6)-Ia, aph (3'')-Ib, qnrS1, oqxA, sul1, dfrA1, tet(A)	fyuA, irp1, irp2, ybtA, ybtE, ybtQ, ybtS, ybtT, ybtU, ybtX, mrkA, mrkB, mrkC, mrkD, mrkF, mrkH, mrkI, mrkJ	R	R	R	R	R	R	R	R	R	R	R
Kp3	Urine	Oct-2015	ST16	K15, K17, K51, K52 O3b	blaCTX-M-15, blaSHV-1, blaOXA-1, blaTEM-1B, blaOXA-181, aac (6')-Ib-cr, aac (3)-IId, aadA5, mph(A), catB3, oqxA, sul1, dfrA17, tet(B)	irp2, iucA, iucC, iucD, iutA, ybtA, ybtE, ybtP, ybtQ, ybtS, ybtT, ybtU, ybtX, traT, fimA, mrkA, mrkB, mrkC, rkD, mrkF, mrkH, mrkI, mrkJ	R	R	R	R	R	R	R	R	R	R	R
Kp4	Urine	Oct-2015	ST147	K64 O2v1	blaCTX-M-15, blaSHV-11, blaOXA-1, blaTEM-1B, blaOXA-181, aac(6')-Ib-cr, aph(3')-Ib, aph(6)-IId, catB3, oqxA, qnrS1, sul1, dfrA1, tet(A)	fyuA, irp1, ybtA, ybtE, ybtQ, ybtS, ybtT, ybtU, ybtX, mrkA, mrkB, mrkC, mrkD, mrkF, mrkH, mrkI, mrkJ	R	R	R	R	R	R	R	R	R	R	R
Kp5	Urine	Nov-2015	ST15	K112 O1v1	blaCTX-M-15, blaSHV-28, blaOXA-1, blaDHA-1, blaLAP-2, aac(6')-Ib-cr, aph(3')-Ia, fosA, catB3, oqxA, qnrS, qnrB4, sul1, sul2, dfrA14, tet(A)	kfuA, kfuB, kfuC, mrkA, mrkB, mrkC	R	R	R	R	R	R	R	R	I	S	S

(continued)

TABLE 1. (CONTINUED)

Molecular typing					Resistance profile CLSI disk diffusion												
Strain ID	Sample	Date	ST	Serotypes	AMR genes	Virulence genes	CPD	CAZ	CTX	CRO	AZM	FEP	FOX	CTT	IPM	MEM	ETP
Kp6	Urine	Nov-2015	ST147	K64 O2v1	<i>bla</i> _{CTX-M-15} , <i>bla</i> _{SHV-11} , <i>bla</i> _{OXA-1} , <i>bla</i> _{TEM-1B} , <i>bla</i> _{OXA-181} , <i>aac</i> (6')- <i>Ib-cr</i> , <i>aac</i> (3)- <i>Ila</i> , <i>aph</i> (3'')- <i>Ib</i> , <i>aph</i> (6)- <i>Id</i> , <i>catB3</i> , <i>oqxA</i> , <i>qnrS1</i> , <i>sul1</i> , <i>dfrA14</i> , <i>tet</i> (A)	<i>fyuA</i> , <i>irp2</i> , <i>ybtA</i> , <i>ybtQ</i> , <i>ybtS</i> , <i>ybtT</i> , <i>ybtU</i> , <i>ybtX</i> <i>mrkA</i> , <i>mrkB</i> , <i>mrkC</i> , <i>mrkD</i> , <i>mrkF</i> , <i>mrkH</i> , <i>mrkI</i> , <i>mrkJ</i>	R	R	R	R	R	R	R	R	R	R	R
Esc	Urine	Oct-2015	ST131	O25:H4	<i>bla</i> _{CTX-M-15} , <i>bla</i> _{TEM-1B} , <i>bla</i> _{OXA-1} , <i>aac</i> (6')- <i>Ib</i> - <i>cr</i> , <i>aph</i> (6)- <i>Id</i> , <i>aac</i> (3)- <i>Ila</i> , <i>aadA5</i> , <i>nph</i> (A), <i>mdf</i> (A), <i>catB3</i> , <i>sul1</i> , <i>sul2</i> , <i>dfrA17</i> , <i>tet</i> (A)	<i>afaA</i> , <i>afaC</i> , <i>afaD</i> , <i>chuA</i> , <i>fyuA</i> , <i>iha</i> , <i>irp2</i> , <i>iss</i> , <i>iucC</i> , <i>iutA</i> , <i>kpsE</i> , <i>kpsMII</i> , <i>ompT</i> , <i>sat</i> , <i>sitA</i> , <i>terC</i> , <i>traT</i> , <i>yfcV</i>	R	R	R	R	R	R	R	R	S	S	R
ECC	Urine	Oct-2015	ST93	—	<i>bla</i> _{ACT-7} , <i>bla</i> _{CTX-M-15} , <i>bla</i> _{TEM-1B} , <i>bla</i> _{OXA-1} , <i>bla</i> _{NDM-4} , <i>bla</i> _{OXA-181} , <i>aac</i> (3)- <i>Ila</i> , <i>aac</i> (6')- <i>Ib-c</i> , <i>aph</i> (3'')- <i>Ib</i> , <i>aph</i> (6)- <i>Id</i> , <i>fosA</i> , <i>nph</i> (A), <i>aac</i> (6')- <i>Ib</i> - <i>cr</i> , <i>sul2</i> , <i>dfrA14</i> , <i>catA2</i> , <i>tet</i> (D)	<i>mrkB</i> , <i>mrkF</i> , <i>fimA</i> , <i>fimD</i> , <i>csgA</i> , <i>csgD</i> , <i>csgE</i> , <i>csgF</i> , <i>acrA</i> , <i>entA</i> , <i>entB</i> , <i>entC</i> , <i>entE</i> , <i>entF</i> , <i>entS</i> , <i>fepA</i> , <i>fepB</i> , <i>fepC</i> , <i>fepD</i> , <i>fepG</i> , <i>T6SS gene</i> <i>cluster</i> , <i>csgB</i> , <i>csgC</i> , <i>csgG</i>	R	R	R	R	R	R	R	R	R	R	R

AMR, antimicrobial resistance; AZM, aztreonam; CAZ, ceftazidime; CPD, cefpodoxime; CRO, ceftriaxone; CTT, cefotetan; CTX, cefotaxime; ECC, *Enterobacter cloacae* complex; ETP, ertapenem; FEP, cefepime; FOX, cefoxitin; I, intermediate; IPM, imipenem; MEM, meropenem; R, resistant; S, susceptible; ST, sequence types.

a globally disseminated cause of urinary tract infections (UTI) and sepsis.⁷ *Enterobacter xiangfangensis* ST114, *Enterobacter hormaechei* subsp. *steigerwaltii* ST90 and ST93, and *E. cloacae* cluster III ST78 are widely disseminated MDR strains of the ECC.² The presence of a variety of antimicrobial resistance (AMR) genes, conferring resistance to several antimicrobial classes, in combination with genes coding for a range of virulence factors, often carried on the same plasmid, enhances the pathogenic potential of these MDR clones.^{8–10}

There have been several reports of outbreaks in Sri Lanka due to MDR *K. pneumoniae* and *E. coli* strains, which resulted in closure of wards.^{11,12} However, except for one, all the other studies conducted on MDR *Enterobacteriales* in the country were based on epidemiological data or Sanger sequencing data.¹³ Studies performing comparative genomics analysis to track the dissemination and transmission of such MDR strains are still limited in the country. The population structure and genetic characteristics of MDR *Enterobacteriales* causing infections are largely unknown. As an attempt to address this knowledge gap, in this study, eight uropathogenic *Enterobacteriales* strains found to be MDR by disk diffusion from hospital-acquired UTI (HA-UTI) were characterized using comparative genomics to identify the STs, serotypes, AMR genes, virulence genes, and associated mobile genetic elements (MGE).

Methods

Study setting

Eight *Enterobacteriales* strains found to be MDR with carbapenem resistant by disk diffusion¹⁴ were randomly selected from a collection of uropathogenic *Enterobacteriales* from HA-UTI from two tertiary care hospitals, Sri Jayawardenepura General Hospital (SJGH) and Neville Fernando Teaching Hospital (NFTH), in the Western Province of Sri Lanka in 2015/2016. Thirty-one isolates of 120 uropathogenic *Enterobacteriales* showed carbapenem resistance by disk diffusion and of these 8 *Enterobacteriales* strains, 6 *K. pneumoniae* strains (Kp1–Kp6), 1 *E. coli* strain (Esc), and 1 *E. cloacae* strain (ECC) were randomly selected to further characterize by whole genome sequencing (WGS) and comparative genome analysis (Table 1). Ethical approval for this study was obtained from the Ethics Review Committee of the University of Colombo, Faculty of Medicine [EC-14-143].

WGS and comparative genome analysis

WGS was performed by the MicrobesNG service. Comparative genome analysis of WGS data was done using bioinformatics tools. The species of the ECC strain and the STs of the eight isolates were identified by the multilocus sequence typing tool provided by the Center for Genomic Epidemiology

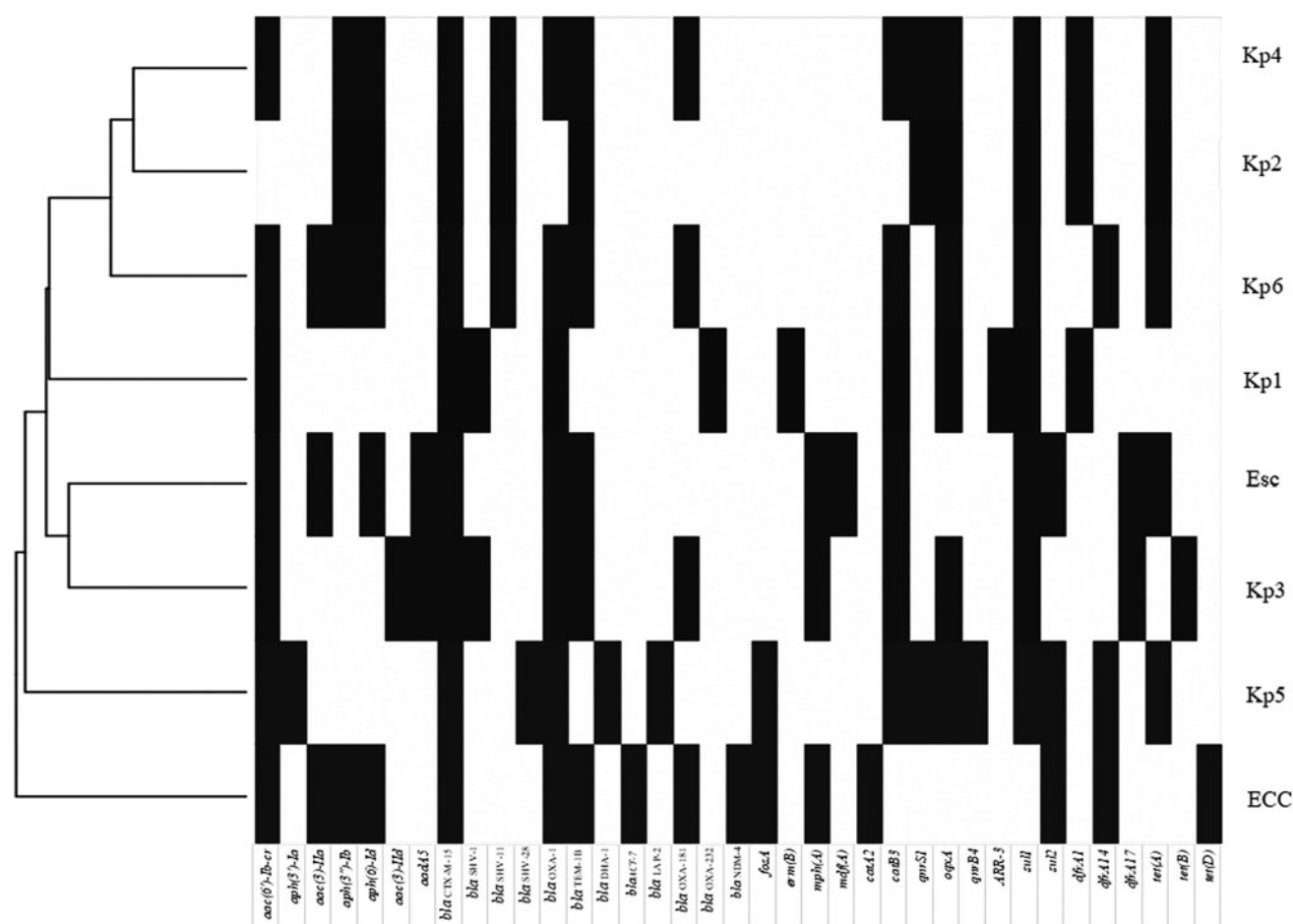


FIG. 1. The dendrogram and heat map generated using the presence and absence of AMR genes (black boxes indicate the presence of genes and white boxes indicate the absence of genes). AMR, antimicrobial resistance.

(CGE)¹⁵ and the Institut Pasteur BIGSdb database.¹⁶ Capsule synthesis (K) and lipopolysaccharide (O) loci of *K. pneumoniae* were determined using Kaptive2.¹⁷ *E. coli* serotype was determined using SerotypeFinder 2.0¹⁸ and the FimH type was identified by FimTyper 1.0.¹⁹

The presence of virulence genes in the *K. pneumoniae* isolates was investigated using the Institut Pasteur BIGSdb database.¹⁶ Virulence genes of *E. coli* were investigated using virulencefinder 2.0 at CGE and the virulence genes of ECC were queried from the virulence factor database using reference genomes.²⁰ The National Center for Biotechnology Information (NCBI) GenBank Basic Local Alignment Search Tool (BLAST) tool²¹ was used to find the corresponding regions with the highest similarity scores (100% coverage, 99% identity) to confirm identification. Antibiotic resistance genes were recognized using Resfinder 2.1 provided by the CGE.²²

MGE flanking AMR genes and virulence genes were identified by MobileElementFinder.²³ Plasmid replicons and MGE were further identified by PlasmidFinder 1.3²⁴ and by ISFinder, respectively.²⁵ The NCBI GenBank BLAST tool²¹ was used to confirm the corresponding regions with the highest similarity scores (100% coverage, 99% identity).²¹ Schematic maps of the plasmids and the flanking regions of the *bla* genes were created using snapgene.

Whole-genome sequences of the eight Sri Lankan *Enterobacteriales* strains have been submitted to the GenBank at NCBI under BioProject accession number PRJNA717825. The genome sequence accession numbers are JAGKSN000000000, JAGKSM000000000, JAGRPG000000000, JAGKSL000000000, JAGKSK000000000, JAGKSJ000000000, JAGKSH000000000, and JAGKSI000000000.

Results

STs and serotypes

The six *K. pneumoniae* isolates belonged to three STs: Kp2, Kp4, and Kp6 were ST147, Kp1 and Kp3 were ST16, and Kp5 was ST15, respectively. Multiple K and O loci were seen in these strains (Table 1). Esc was identified as *E. coli* ST131 belonging to O25:H4 serotype. The single ECC isolate was identified as *E. hormaechei* subsp. *steigerwaltii* ST93 (Table 1).

AMR genes

A total of 37 AMR genes conferring resistance to different antimicrobial classes were identified with all isolates harboring genes coding for resistance to six or more antibiotic classes (Fig. 1). Resistance genes to aminoglycosides [*aac*(6′)-Ib-cr, *aph*(3′)-Ia, *aac*(3)-IIa, *aph*(3′′)-Ib, *aph*(6)-Id, *aac*(3)-IIc, *aadA5*], β-lactams [*bla*_{CTX-M-15}, *bla*_{SHV-1}, *bla*_{SHV-11}, *bla*_{SHV-28}, *bla*_{TEM-1B}, *bla*_{OXA-1}, *bla*_{DHA-1}, *bla*_{ACT-7}, *bla*_{LAP-2}, *bla*_{OXA-181}, *bla*_{OXA-232}, *bla*_{NDM-4}], fosfomycin (*fosA*), macrolides [*erm*(B), *mph*(A), *mdh* (A)], chloramphenicol (*catA2*, *catB3*), sulfonamides (*sul1*, *sul2*), trimethoprim (*dfrA1*, *dfrA14*, *dfrA17*), quinolones [*aac*(6′)-Ib-cr, *oqxA*, *qnrS1*, *qnrB4*], rifampicin (*ARR-3*), and tetracycline [*tet*(A), *tet*(B), *tet*(D)] was seen.

β-Lactamase genes

Kp3, Kp4, and Kp6 harbored *bla*_{OXA-181}, whereas Kp1 harbored *bla*_{OXA-232}. Kp5 carried multiple ESBL and AmpC β-lactamase genes *bla*_{CTX-M-15}, *bla*_{SHV-28}, *bla*_{OXA-1}, and *bla*_{DHA-1}. This isolate also carried the acquired narrow

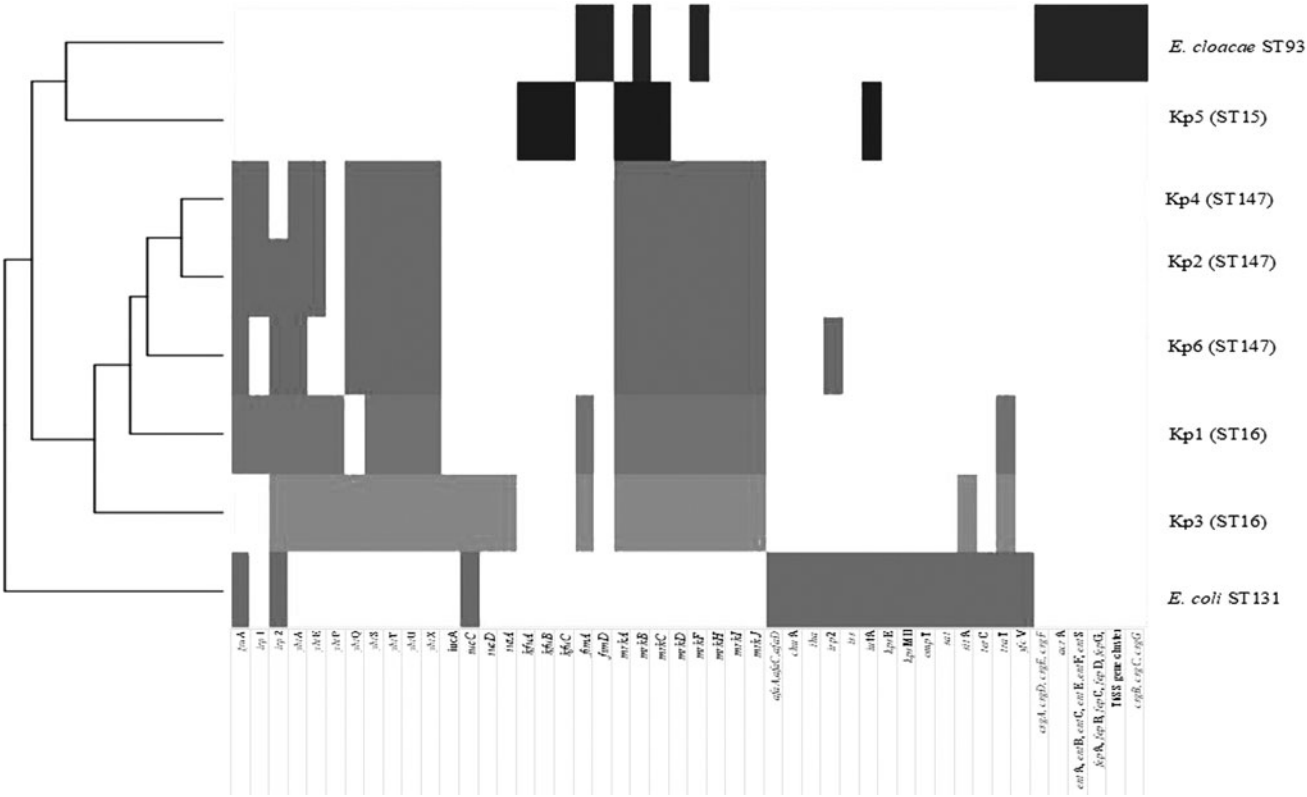


FIG. 2. The dendrogram and heat map generated using the presence and absence of virulence genes (black boxes indicate the presence of genes and white boxes indicate the absence of genes).

spectrum *bla*_{LAP-2} gene. Kp2 carried *bla*_{CTX-M-15}, *bla*_{SHV-11}, and *bla*_{TEM-1B}, and Esc carried *bla*_{CTX-M-15}, *bla*_{OXA-1}, and *bla*_{TEM-1B} (Table 1 and Fig. 1). The ECC strain carried both *bla*_{OXA-181} and *bla*_{NDM-4} together with the *ble*_{MBL} gene, which encodes resistance to bleomycin.

Virulence genes

Virulence genes coding for several functional proteins were identified (Fig. 2). Siderophore genes: *fyuA*, *irp1*,

irp2, *iucA*, *iucD*, *iutA*, *ybtA*, *ybtE*, *ybtP*, *ybtQ*, *ybtS*, *ybtT*, *ybtU*, *ybtX*, *chuA*, *fepA*; iron uptake genes: *kfuA*, *kfuB*, *kfuC*, *irp*, *sitA*; fimbrial adherence determinant genes: *fimA*, *fimD*, *mrkA*, *mrkB*, *mrkC*, *mrkD*, *mrkF*, *mrkH*, *mrkI*, *mrkJ*, *afaA*, *afaC*, *afaD*, *iha*, *yfcV*; serum survival gene: *iss*; capsular polysaccharide gene: *kps*; outer membrane protease: *ompT*; secreted autotransporter toxin: *sat*; tellurium ion resistance protein: *terC*; serum resistance-associated outer membrane lipoprotein: *traT*; Curli fiber gene: *csgA*, *csgB*, *csgC*, *csgD*, *csgE*, *csgF*, *csgG*; efflux

TABLE 2. MOBILE GENETIC ELEMENTS ASSOCIATED WITH ANTIMICROBIAL RESISTANCE AND VIRULENCE GENES

	AMR gene	Virulence gene	Plasmid replicons	MGE
Kp1 (ST16)	<i>erm(B)</i> <i>bla</i> _{CTX-M-15} <i>bla</i> _{OXA-1-catB3-ARR3-emrE-sul1 <i>bla</i>_{OXA-232}}	<i>traT</i>	ColKP3 IncFII(K)	IS26-IS26 composite transposon ISEcp1, Tn3-like transposon Tn3-like transposon ISEcp1
Kp2 (ST147)	<i>bla</i> _{CTX-M-15} <i>sul1</i>	<i>fyuA</i>	IncR	ISEcp1, Tn3-like transposon IS6100 ISEc33
Kp3 (ST16)	<i>bla</i> _{CTX-M-15} <i>aac(6')-Ib-cr-bla</i> _{OXA-1-catB3} <i>bla</i> _{TEM-1B} <i>bla</i> _{OXA-181}	<i>traT</i> <i>sitA</i>	IncFII ColKP3 IncFIA	ISEcp1, Tn3-like transposon ISEc9, IS26-IS26 composite transposon ISEcp1 ISKpn8
Kp4 (ST147)	<i>bla</i> _{CTX-M-15} <i>aac(6')-Ib-cr-bla</i> _{OXA-1-catB3} <i>tet(A)</i>		IncR	ISEcp1, Tn3-like transposon IS26-IS26 composite transposon IS6100
Kp5 (ST15)	<i>bla</i> _{CTX-M-15} <i>aac(6')-Ib-cr-bla</i> _{OXA-1-catB-sul} <i>aph(3')-Ia</i> <i>qnrS1</i> <i>bla</i> _{Lap-2}	<i>iutA</i>		ISEcp1, Tn3-like transposon ISEc33, ISVsa3 IS26-IS26 composite transposon ISKpn19, ISEcl1 ISEcl1
Kp6 (ST147)	<i>bla</i> _{CTX-M-15} <i>aac(6')-Ib-cr-bla</i> _{OXA-1-catB3} <i>tet(A)</i> <i>bla</i> _{TEM-1B} <i>bla</i> _{OXA-181}	<i>irp2</i>	ColKP3	ISEcp1, Tn3-like transposon IS26-IS26 composite transposon IS6100 ISKpn19 ISEc33 ISEcp1
Esc	<i>bla</i> _{CTX-M-15} <i>aac(6')-Ib-cr-bla</i> _{OXA-1-catB3} <i>dfrA17</i> <i>mdf(A)</i> <i>tet(A)</i> <i>bla</i> _{TEM-1B}	<i>ompT</i> <i>sitA</i> <i>traT</i> <i>iutA</i> <i>fyuA</i> <i>yfcV</i> <i>kpsMII</i> <i>afaC</i> <i>Usp</i> <i>terC</i>	IncFII	ISEcp1, Tn3-like transposon IS26-IS26 composite transposon IS6100 MITEEc1/IS630 Tn5403, Unit transposon ISEc63, Tn3-like transposon IS30, IS629 ISEc38 MITEEc1/IS630 IS4 IS640/IS21 ISEc53 and ISEc52 MITEEc1/IS630
ECC	<i>bla</i> _{CTX-M-15} <i>aac(6')-Ib-cr-bla</i> _{OXA-1-catB3} <i>sul2</i> <i>bla</i> _{NDM-4-ble}			ISEcp1, Tn3-like transposon IS26-IS26 composite transposon IS5075/IS110 IS5

MGE, mobile genetic elements.

transporter periplasmic adaptor subunit: *acrA*; bacteriocins genes: *entA*, *entB*, *entC*, *entE*, *entF*, *entF*; Type VI secretion systems locus: T6SS.

Kp1–Kp4 and Kp6 exhibited the yersiniabactin locus (*ybt*) encoding yersiniabactin siderophores, whereas Kp5 manifested the *kfu* iron uptake system gene. Although all *K. pneumoniae* strains harbored the Type 2 fimbrial gene cluster *mrk*, Kp1 and Kp3 showed additional *fimA* genes. The serum resistance-associated outer membrane lipoprotein *traT* gene was seen in *K. pneumoniae* Kp1 and Kp3 strains and in the *Esc* strain. Genes coding for adherence factors (*afa*, *iha*, *yfcV*), toxins (*sat*), and iron acquisition systems (*iutA*, *sitA*, *chuA*) and several other virulence genes

were present in the *Esc* strain. The curli fiber gene cluster *csg*, efflux transporter periplasmic adaptor subunit *acrA*, bacteriocins gene cluster *ent*, and Type VI secretion systems locus (T6SS) were found in the ECC strain (Table 1 and Fig. 2).

AMR and virulence genes associated MGE

Several MGEs associated with AMR and virulence genes (Table 2). In all isolates *ISEcp1* flanked upstream of *bla*_{CTX-M-15}. In Kp1, Kp3, and Kp6, ColKP3 (size ~6.1 kb) plasmids carried the *bla*_{OXA-48-like} genes flanked by *ISEcp1* (Fig. 3). In Kp2 and Kp4, IncR plasmids (size ~31 kb and ~42 kb) carried *sulI* and *tet(A)* flanked by IS6100

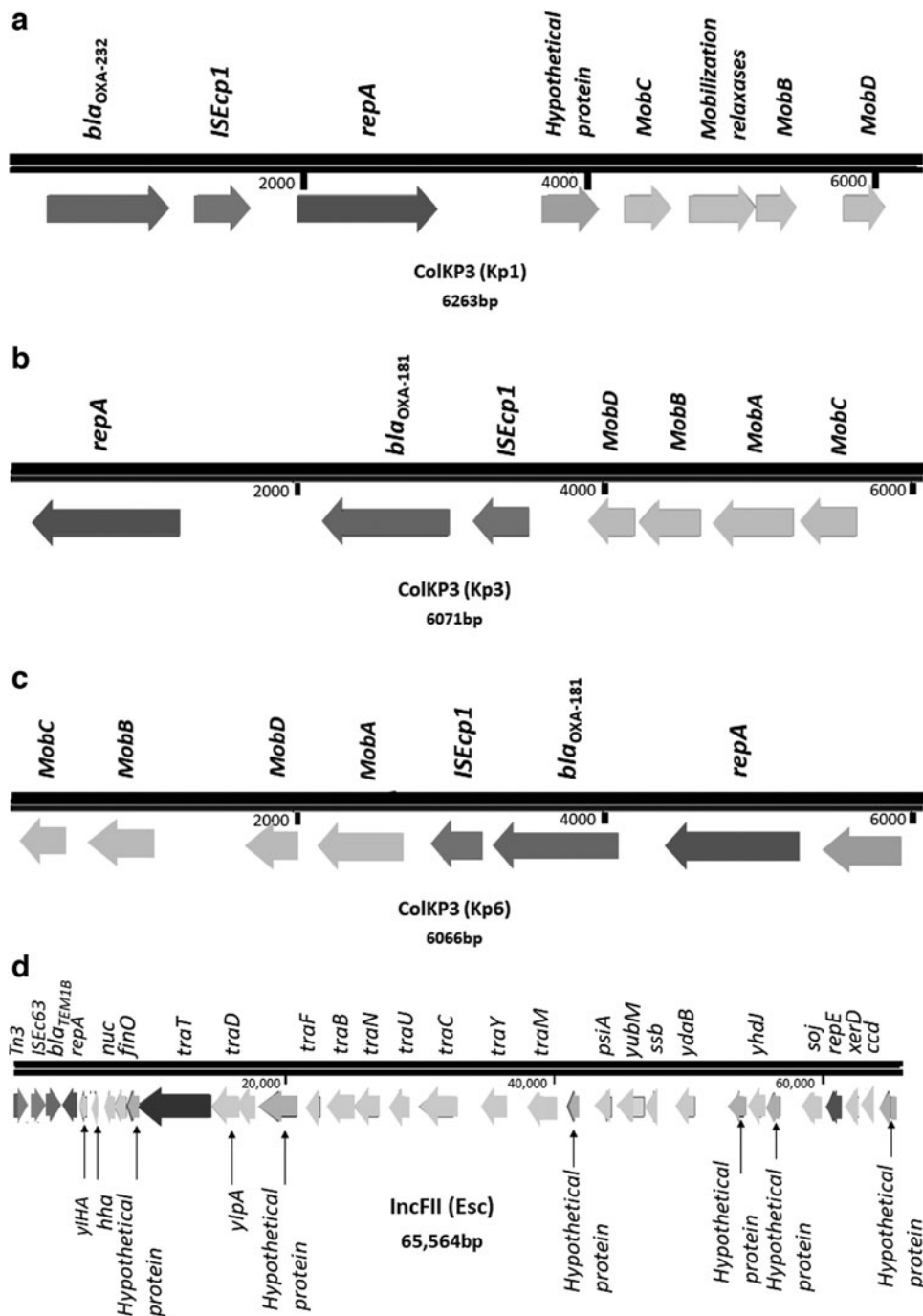


FIG. 3. Plasmid maps of AMR plasmids. **(a)** ColKP3 carrying *bla*_{OXA-232} in Kp1 **(b)** ColKP3 carrying *bla*_{OXA-181} in Kp3 **(c)** ColKP3 carrying *bla*_{OXA-181} in Kp6 **(d)** IncF plasmid carrying *bla*_{TEM-1B} and *traT*.

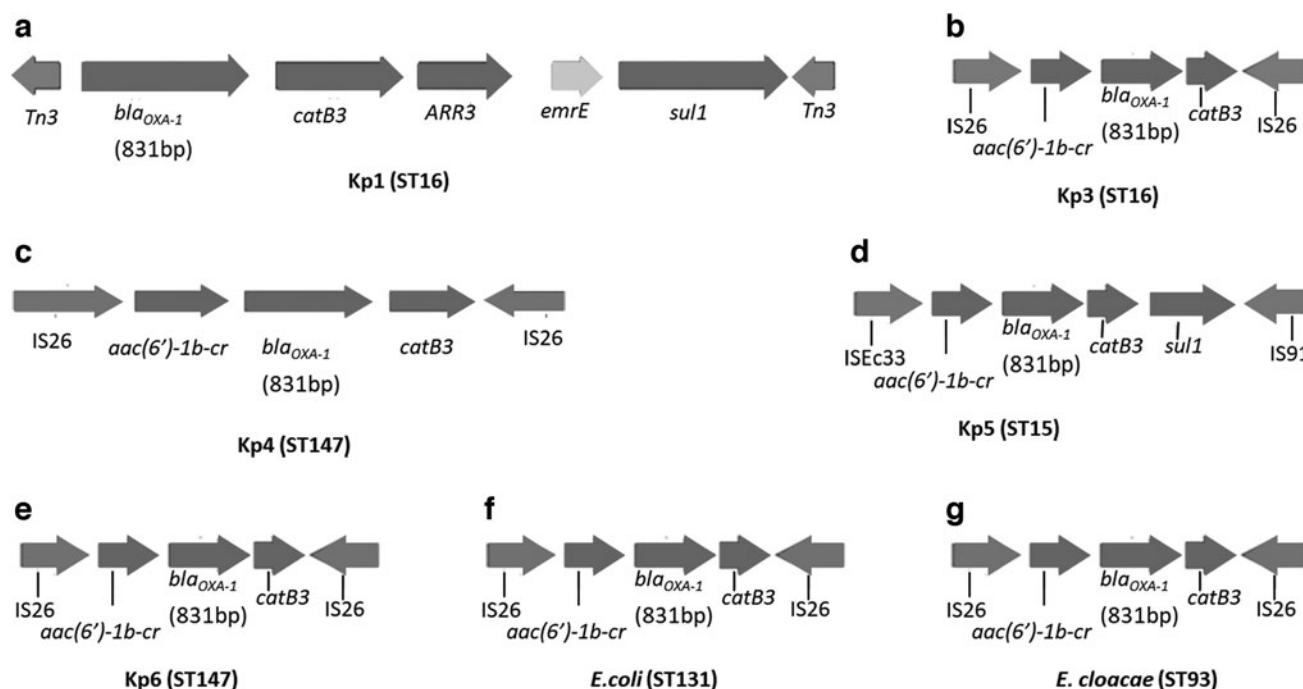


FIG. 4. Clusters of AMR genes associated with MGE: (a) *bla*_{OXA-1}, *catB3*, *ARR3*, *emrE* multidrug-transporter, and *sul1* associated with *Tn3* in Kp1 (b) *aac*(6′)-Ib-cr, *bla*_{OXA-1}, *catB3* associated with *IS26* in Kp3 (c) *aac*(6′)-Ib-cr, *bla*_{OXA-1}, *catB3* associated with *IS26* in Kp4 (d) *aac*(6′)-Ib-cr, *bla*_{OXA-1}, *catB3*, *sul1* associated with *ISEc33*, and *IS91* in Kp5 (e) *aac*(6′)-Ib-cr, *bla*_{OXA-1}, *catB3* associated with *IS26* in Kp6 (f) *aac*(6′)-Ib-cr, *bla*_{OXA-1}, *catB3* associated with *IS26* in Esc (g) *aac*(6′)-Ib-cr, *bla*_{OXA-1}, *catB3* associated with *IS26* in ECC. ECC, *Enterobacter cloacae* complex.

(Table 2). In Kp1, Kp3, and Esc, IncF plasmids (size ~59 kb) carried virulence genes (e.g., *traT*, *sitA*). Furthermore, in Kp1 and Esc, the *traT* virulence gene was carried together with the AMR gene *bla*_{TEM-1B} on IncF (Fig. 3). Clusters of AMR genes, encoding resistance to multiple antimicrobial classes associated with MGE were identified in seven of the eight strains (Kp1, Kp3–Kp6, Esc, and ECC) (Fig. 4).

Discussion

We studied six MDR *K. pneumoniae* (Kp1–Kp6), one *E. coli* (Esc), and one *E. cloacae* (ECC) strain, using comparative genomics. Although *K. pneumoniae* ST15 and ST147 have been reported in Sri Lanka previously,^{26,27} this article describes the isolation of *K. pneumoniae* ST16, *E. coli* ST131, and *E. hormaechei* subsp. *steigerwaltii* ST93 for the first time in Sri Lanka. These STs are globally successful clones, known to disseminate MDR and associated with hospital outbreaks in many countries and are reported globally disseminated cause of difficult to treat UTIs and sepsis.^{2,5–7}

The isolates harbored genes coding for resistance to six or more antibiotic classes exemplifying co-selection of resistance genes. The co-occurrence of multiple AMR genes in Sri Lanka has been reported previously only in *K. pneumoniae*.¹⁴ This article broadens this to include both *E. coli* and *E. hormaechei*. These strains confer broad resistance to extended-spectrum beta-lactams, fluoroquinolones, and to carbapenems, which is a serious concern for public health.

Although all the isolates were carbapenem resistant by disk diffusion, three strains, Kp2, Kp5, and Esc, did not

carry any known carbapenemase genes. The combination of multiple Amp C/ESBL production and other resistance mechanisms such as porin deficiency may have resulted in phenotypic carbapenem resistance in these isolates.²⁸ The *bla*_{NDM} gene in the ECC strain was found together with the *ble*_{MBL} gene, which confers resistance to bleomycin, an anticancer agent used in clinical therapy. This association has been identified regularly in *bla*_{NDM}-producing isolates of *Enterobacteriales*.²⁹ Both the *bla*_{NDM} and the *ble*_{MBL} gene are believed to have originated from the same progenitor.³⁰ Such strains harboring both *bla*_{NDM} and the *ble*_{MBL} gene could be selected by antibiotics or co-selected by bleomycin.^{29,31}

In Kp5, *bla*_{LAP-2} and *qnrS1* were found together, flanked by *ISEc1*. However, *ISEc1* suggested no role for *qnrS1* mobilization. *bla*_{LAP-2}-*qnrS1* association flanked by *ISEc1* has been found on unknown plasmids that results their mobilization in *Enterobacteriales* in several countries such as China, Tunisia, Norway, and Netherlands. In our study also, the *bla*_{LAP-2}-*qnrS1* and the associated *ISEc1* were carried on an untypable plasmid.^{32,26}

The *K. pneumoniae* strains harbored fimbrial genes (*fim*, *mrk*) that play a role in the pathogenesis of UTI. The type 3 fimbrial adhesins gene cluster *mrk*, which is associated with binding to the extracellular matrix and biofilm formation, was present in all the *K. pneumoniae* strains. This may play a major role in colonization and invasion of these strains. ESBL-producing isolates of *K. pneumoniae* are reported to have the ability to produce more fimbrial adhesins.²⁷ However, *rmpA* genes associated with the hypervirulent

(hypermucoviscous) variant of *K. pneumoniae* (hvKP) were not identified in our isolates, which suggests that the isolates included in the study do not have the molecular characteristics of the hypervirulent (hypermucoviscous) *K. pneumoniae*.²⁷

In Kp1, Kp3, and Esc the serum resistance-associated outer membrane lipoprotein (*traT*) that has a predominant role in pathogenicity was identified. *TraT* gene reported to be the most prevalent virulence gene among uropathogenic *E. coli* strains, which could be useful as a target for therapeutic interventions. Several extraintestinal pathogenic *E. coli* (ExPEC)-associated virulence genes (*afaA*, *afaC*, *afaD*, *chuA*, *irp2*, *iutA*, *kpsM*, *ompT*, *sitA*, *terC*, *traT*, *yfcV*, *iha*, and *iss*) previously described in highly pathogenic *E. coli* isolated from HA infections in different countries were identified in Esc.³³ Overexpression of the efflux pump gene *acrA* has been observed to increase antibiotic resistance and virulence in *E. cloacae*.³⁰ The T6SS gene cluster coding for the type VI secretory system, which increases the ability of pathogens to interact with a variety of abiotic and biotic surfaces and confer their survival in a range of environments, was also identified in the ECC strain.³⁴

The *bla*_{CTX-M-15} gene in all the isolates had the globally disseminated *ISEcp1-bla*_{CTX-M} type genetic context.³⁵ The *ISEcp1* has been found to facilitate the dissemination of *bla*_{CTX-M-15}-associated MDR determinants among *Enterobacteriales* species. *Bla*_{OXA-48-like} carbapenemase genes (*bla*_{OXA-181} and *bla*_{OXA-232}) in three of our isolates (Kp1, Kp3, and Kp6) were carried by ColKP3 plasmids and were associated with *ISEcp1* that promotes rapid dissemination of carbapenem resistance. This association has been reported previously both globally and locally, which is a serious public health concern.^{13,36,37} Although plasmids of the IncF group are some of the most frequent resistance plasmids found in *Enterobacteriales*, none of the *K. pneumoniae* strains in our study carried AMR genes on IncF plasmid replicons.³⁸

However, these plasmids carried virulence genes such as *traT* and *sitA* in some strains (Kp1 and Kp3). In Esc an AMR gene (*bla*_{TEM-1B}) and a virulence gene (*traT*) were found in combination on an IncF plasmid replicon. High rates of IncF plasmids co-harboring resistant and virulence genes have been reported in China.³⁹ In Kp2, Kp4, and Kp6 the AMR genes present on IncR plasmid replicons (*sul1*, *tetB*) were associated with insertion sequence IS6100. Although IncR plasmids are believed to be nontransferable, association with such transposable elements may allow the spread of these genes.⁴⁰

We found that AMR gene clusters of three to four genes conferring resistance to multiple antimicrobial classes flanked by mobile elements in almost all the strains that may indicate co-transmission of these genes. The most common AMR genes located in these clusters were *aac*(6')-Ib-cr, *bla*_{OXA-1}, and *catB3* flanked by IS26 (composite transposon). This cassette is widely distributed in *Enterobacteriales* in China.⁴¹ IS26 is reported to result in various forms and horizontal spread of AMR.

Conclusion

The concomitant presence of resistance against multiple antibiotics and virulence factors facilitates bacterial patho-

genicity in HA infections. We studied eight MDR uropathogenic *Enterobacteriales* isolated from HA-UTI in Sri Lanka, using genomic sequencing and comparative genome analysis. This study identified a diverse variety of antibiotic resistance genes, virulence genes, and associated MGE that indicate the presence of highly adapted strains of MDR *Enterobacteriales* in the hospital setting and adds to the currently scarce data on MDR uropathogenic *Enterobacteriales* in Sri Lanka.

Disclosure Statement

No competing financial interests exist.

Funding Information

This work was supported by grants from the National Research Council [Grant number 14-45] and the University of Colombo [Grant number AP/3/2/2018/SG/16].

References

1. Neubeiser A, Bonsignore M, Tafelski S, et al. Mortality attributable to hospital acquired infections with multidrug-resistant bacteria in a large group of German hospitals. *J Infect Public Health* 2020;13(2):204–210; doi: 10.1016/j.jiph.2019.07.025.
2. Annavajhala MK, Gomez-Simmonds A, Uhlemann AC. Multidrug-resistant *Enterobacter cloacae* complex emerging as a global, diversifying threat. *Front Microbiol* 2019; 10:44; doi: 10.3389/fmicb.2019.00044.
3. Chen LF, Freeman JT, Nicholson B, et al. Widespread dissemination of CTX-M-15 genotype extended-spectrum-β-lactamase-producing *Enterobacteriaceae* among patients presenting to community hospitals in the southeastern United States. *Antimicrob Agents Chemother* 2014;58(2): 1200–1202; doi: 10.1128/AAC.01099-13.
4. Sabirova JS, Xavier BB, Coppens J, et al. Whole-genome typing and characterization of *bla* VIM19-harboring ST383 *Klebsiella pneumoniae* by PFGE, whole-genome mapping and WGS. *J Antimicrob Chemother* 2016;71(6): 1501–1509; doi: 10.1093/jac/dkw003.
5. Espinal P, Nucleo E, Caltagirone M, et al. Genomics of *Klebsiella pneumoniae* ST16 producing NDM-1, CTX-M-15, and OXA-232. *Clin Microbiol Infect* 2019;25(3):385–e1; doi: 10.1016/j.cmi.2018.11.004.
6. Wyres KL, Nguyen TN, Lam MM, et al. Genomic surveillance for hypervirulence and multi-drug resistance in invasive *Klebsiella pneumoniae* from South and Southeast Asia. *Genome Med* 2020;12(1):1–6; doi: 10.1186/s13073-019-0706-y.
7. Petty NK, Zakour NL, Stanton-Cook M, et al. Global dissemination of a multidrug resistant *Escherichia coli* clone. *Proc Natl Acad Sci U S A* 2014;111(15):5694–5699. doi: 10.1073/pnas.1322678111.
8. Roe CC, Vazquez AJ, Esposito EP, et al. Diversity, virulence, and antimicrobial resistance in isolates from the newly emerging *Klebsiella pneumoniae* ST101 lineage. *Front Microbiol* 2019;10:542. doi: 10.3389/fmicb.2019.00542.
9. Ireng LM, Ambroise J, Bearzatto B, et al. Whole-genome sequences of multidrug-resistant *Escherichia coli* in South-Kivu Province, Democratic Republic of Congo: Characterization of phylogenomic changes, virulence and resistance genes. *BMC Infect Dis* 2019;19(1):1–0. doi: 10.1186/s12879-019-3763-3.

10. Fekete PZ, Brzuszkiewicz E, Blum-Oehler G, et al. DNA sequence analysis of the composite plasmid pTC conferring virulence and antimicrobial resistance for porcine enterotoxigenic *Escherichia coli*. *Int J Med Microbiol* 2012; 302(1):4–9; doi: 10.1016/j.ijmm.2011.07.003.
11. Senanayake N, Karunanayake L. Outbreak of bloodstream infection with extended spectrum β -lactamase-producing *Klebsiella pneumoniae* at a Teaching Hospital. *J Nepal Paediatr Soc* 2015;34(3):230–232; doi: 10.3126/jnps.v34i3.10459.
12. Tissera K, Liyanapathirana V, Dissanayake N, et al. Spread of resistant gram negatives in a Sri Lankan intensive care unit. *BMC Infect Dis* 2017;17(1):1–0; doi: 10.1186/s12879-017-2590-7.
13. Zhu C, Liyanapathirana V, Li C, et al. Characterizing mobilized virulence factors and multidrug resistance genes in carbapenemase-producing *Klebsiella pneumoniae* in a Sri Lankan Hospital. *Front Microbiol* 2018;9:2044; doi: 10.3389/fmicb.2018.02044.
14. Clinical and Laboratory Standards Institute 2018. Performance Standards for Antimicrobial Disk Susceptibility Tests, 13th ed. CLSI standard M02. Clinical and Laboratory Standards Institute: Wayne, PA.
15. Larsen MV, Cosentino S, Rasmussen S, et al. Multilocus sequence typing of total-genome-sequenced bacteria. *J Clin Microbiol* 2012;50(4):1355–1361; doi: 10.1128/JCM.06094-11.
16. Jolley KA, Bray JE, Maiden MC. Open-access bacterial population genomics: BIGSdb software, the PubMLST.org website and their applications. *Wellcome Open Res* 2018;3; doi: 10.12688/wellcomeopenres.14826.1.
17. Wick RR, Heinz E, Holt KE, et al. Kaptive Web: User-friendly capsule and lipopolysaccharide serotype prediction for *Klebsiella* genomes. *J Clin Microbiol* 2018;56(6); doi: 10.1128/JCM.00197-18.
18. Joensen KG, Scheut F, Lund O, et al. Real-time whole-genome sequencing for routine typing, surveillance, and outbreak detection of verotoxigenic *Escherichia coli*. *J Clin Microbiol* 2014;52(5):1501–1510; doi: 10.1128/JCM.03617-13.
19. Roer L, Tchesnokova V, Allesøe R, et al. Development of a web tool for *Escherichia coli* subtyping based on fimH alleles. *J Clin Microbiol* 2017;55(8):2538–2543; doi: 10.1128/JCM.00737-17.
20. Chen L, Yang J, Yu J, et al. VFDB: a reference database for bacterial virulence factors. *Nucleic Acids Res* 2005; 33(suppl_1):D325–D328; doi: 10.1093/nar/gki008.
21. Johnson M, Zaretskaya I, Raytselis Y, et al. NCBI BLAST: a better web interface. *Nucleic Acids Res* 2008; 36(suppl_2):W5–W9; doi: 10.1093/nar/gkn201.
22. Bortolaia V, Kaas RS, Ruppe E, et al. ResFinder 4.0 for predictions of phenotypes from genotypes. *J Antimicrob Chemother* 2020;75(12):3491–3500; doi: 10.1093/jac/dkaa345.
23. Johansson M, Bortolaia V, Tansirichaiya S, et al. Detection of mobile genetic elements associated with antibiotic resistance in *Salmonella enterica* using a newly developed web tool: MobileElementFinder. *J Antimicrob Chemother* 2020;76(1):101–109; doi: 10.1093/jac/dkaa390.
24. Carattoli A, Zankari E, García-Fernández A, et al. In silico detection and typing of plasmids using PlasmidFinder and plasmid multilocus sequence typing. *Antimicrob Agents Chemother* 2014;58(7):3895–3903; doi: 10.1128/AAC.02412-14.
25. Siguier P, Pérochon J, Lestrade L, et al. ISfinder: The reference centre for bacterial insertion sequences. *Nucleic Acids Res* 2006;34(suppl_1):D32–D36; doi: 10.1093/nar/gkj014.
26. Poirer L, Leviandier C, Nordmann P. Prevalence and genetic analysis of plasmid-mediated quinolone resistance determinants QnrA and QnrS in Enterobacteriaceae isolates from a French university hospital. *Antimicrob Agents Chemother* 2006;50(12):3992–3997; doi: 10.1128/AAC.00597-06aa.
27. Ferreira RL, da Silva B, Rezende GS, et al. High prevalence of multidrug-resistant *Klebsiella pneumoniae* harboring several virulence and β -lactamase encoding genes in a Brazilian intensive care unit. *Front Microbiol* 2019;22: 9:3198.
28. Sugawara E, Kojima S, Nikaido H. *Klebsiella pneumoniae* major porins OmpK35 and OmpK36 allow more efficient diffusion of β -lactams than their *Escherichia coli* homologs OmpF and OmpC. *J Bacteriol* 2016;198(23): 3200–3208.
29. Poirer L, Dortet L, Bernabeu S, et al. Genetic features of blaNDM-1-positive Enterobacteriaceae. *Antimicrob Agents Chemother* 2011;55(11):5403–5407; doi: 10.1128/AAC.00585-11.
30. Telke AA, Olaitan AO, Morand S, et al. soxRS induces colistin hetero-resistance in *Enterobacter asburiae* and *Enterobacter cloacae* by regulating the acrAB-tolC efflux pump. *J Antimicrob Chemother* 2017;72(10):2715–2721.
31. Zhai Y, Lee S, Teng L, et al. “Dissemination mechanisms of NDM genes in hospitalized patients.” *JAC-Antimicrob Resist* 2021;3(1):dlab032.
32. Kocsis B, Kocsis E, Fontana R, et al. Identification of blaLAP-2 and qnrS1 genes in the internationally successful *Klebsiella pneumoniae* ST147 clone. *J Med Microbiol* 2013;62(2):269–273; doi: 10.1099/jmm.0.050542-0.
33. Ranjbar R, Kelishadrokhi AF, Chehelgerdi M. Molecular characterization, serotypes and phenotypic and genotypic evaluation of antibiotic resistance of the *Klebsiella pneumoniae* strains isolated from different types of hospital-acquired infections. *Infect Drug Resist* 2019;12: 603.
34. Mustafa A, Ibrahim M, Rasheed MA, et al. Genome-wide analysis of four *Enterobacter cloacae* complex type strains: Insights into virulence and niche adaptation. *Sci Rep* 2020;10(1):1–9; doi: 10.1038/s41598-020-65001-4.
35. Singh NS, Singhal N, Virdi JS. Genetic environment of blaTEM-1, blaCTX-M-15, blaCMY-42 and characterization of integrons of *Escherichia coli* isolated from an Indian urban aquatic environment. *Front Microbiol* 2018;9:382; doi: 10.3389/fmicb.2018.00382.
36. Lutgring JD, Zhu W, de Man TJ, et al. Phenotypic and genotypic characterization of Enterobacteriaceae producing oxacillinase-48-like carbapenemases, United States. *Emerg Infect Dis* 2018;24(4):700; doi: 10.3201/eid2404.171377.
37. Findlay J, Hopkins KL, Loy R, et al. OXA-48-like carbapenemases in the UK: An analysis of isolates and cases from 2007 to 2014. *J Antimicrob Chemother* 2017;72(5): 1340–1349; doi: 10.1093/jac/dkx012.
38. Navon-Venezia S, Kondratyeva K, Carattoli A. *Klebsiella pneumoniae*: A major worldwide source and shuttle for antibiotic resistance. *FEMS Microbiol Rev* 2017;41(3): 252–275; doi: 10.1093/femsre/fux013.

39. Yang Y, Zhang A, Lei C, et al. Characteristics of plasmids coharboring 16S rRNA methylases, CTX-M, and virulence factors in *Escherichia coli* and *Klebsiella pneumoniae* isolates from chickens in China. *Foodborne Pathog Dis* 2015;12(11):873–880; doi: 10.1089/fpd.2015.2025.
40. Compain F, Frangeul L, Drieux L, et al. Complete nucleotide sequence of two multidrug-resistant IncR plasmids from *Klebsiella pneumoniae*. *Antimicrob Agents Chemother* 2014;58(7):4207–4210; doi: 10.1128/AAC.02773-13.
41. Tao Y, Zhou K, Xie L, et al. Emerging coexistence of three PMQR genes on a multiple resistance plasmid with a new surrounding genetic structure of qnrS2 in *E. coli* in China. *Antimicrob Resist Infect Control* 2020;9(1):1–8; doi: 10.1186/s13756-020-00711-y.

Address correspondence to:
 Vindya Perera, BVSc
 Department of Microbiology
 Faculty of Medicine
 University of Colombo
 Colombo 00800
 Sri Lanka

E-mail: v.perera83@yahoo.com